

Project: PickTick
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1.0	03/02/2026	Team 1	Initial Draft	

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1. Introduction and Project Overview

The purpose of this project is to develop a mobile marketplace called PickTick where users can buy and resell tickets for events such as concerts, sports games, theater shows, or festivals. The primary problem this system solves is the lack of structure and the insecurity that other ticket-selling platforms provide, by ensuring sellers' information, ticket proof for both physical and digital form, and fraud detecting by admin after carefully review user report.

The application support 3 main roles, include buyer, seller and administrator, each with clearly responsibility. Buyers can browse available tickets, search and filter by event details, view ticket information, and complete checkout using a mock payment flow. Sellers can list tickets they own and manage their sales, as long as they can give ticket information and ticket proof. Administrators can review users account, checking report and ticket listings to prevent fraud and ensure platform safety. The system will focus on structured ticket management, secure clearly transaction and moderated communication between users.

Major System Features

- **User Registration:** Allow new users to create an account by entering user information and selecting their role (buyer/user – this can also be changed later).
- **User Login:** User must log in before starting to use the feature in the app. This ensures account security and personal confirmations.
- **Buyer Module:** Buyer can browse existing listing tickets, search for events, view ticket details, add tickets to cart, and place orders.
- **Seller Module:** Seller can upload their ticket, post ticket details and proof, create a price, and view their ticket status
- **Admin Module:** Administrator can review user accounts and listings, handling user reports to take appropriate action to prevent scam and fraud.
- **Marketplace:** The page where buyers look for their desired ticket, and sellers post what they want to sell. Here, each ticket will show its event name, time, place, and price.
- **Search and Sort:** Make it easier for users to navigate what ticket they demand by searching the event name or filtering the date, location, categories, price range, etc.
- **Payment Processing:** Handle financial transactions to simulate secure and transparent checkout between buyer and seller.
- **Chatbox:** A place where users can communicate with each other, especially for sellers and buyers to message each other and negotiate to get the ideal price for both

2. Objectives

2.1 BUSINESS OBJECTIVES

1. **Registration:** The system will ask for some personal information to verify for newuser account, this include:
 - First Name, Middle Name(optional), Last Name
 - Phone Number(optional)
 - Address
 - Email Address
 - Role Selection(Buyer/Seller)
 - Username
 - Login ID
 - Password
 - Profile Picture(optional)

The role the user chooses at this step will make it easier for them to capture the main action for each role, either user or buyer, which can be changed later.
2. **Login:** All users have to log in using the information they registered during the registration step to ensure the credential authentication. If the user doesn't have an account yet, it will lead back to Registration
 - Login ID
 - Password
 - Login and Cancel button
 - New User? -> Registration
3. **Search and Filter:** Users can use a search bar to type a full or partial keyword of at least 3 characters long; the result will be related to:
 - Event Name
 - Seller Username
 - Location

Users can also use a filter to sort the results, which includes:

 - Event category
 - Date range
 - Ticket price
 - Location
4. **Ticket Listing:** The system support seller to create a listing for the ticket, which should include:
 - Ticket Name
 - Event Name
 - Date/Time
 - Location
 - Price
 - Category
 - Seat Ticket
 - Proof (picture if physical, confirmation if digital)
5. **Checkout Payment:** The system must provide a checkout platform that enables the user to order a ticket and get a confirmation message:

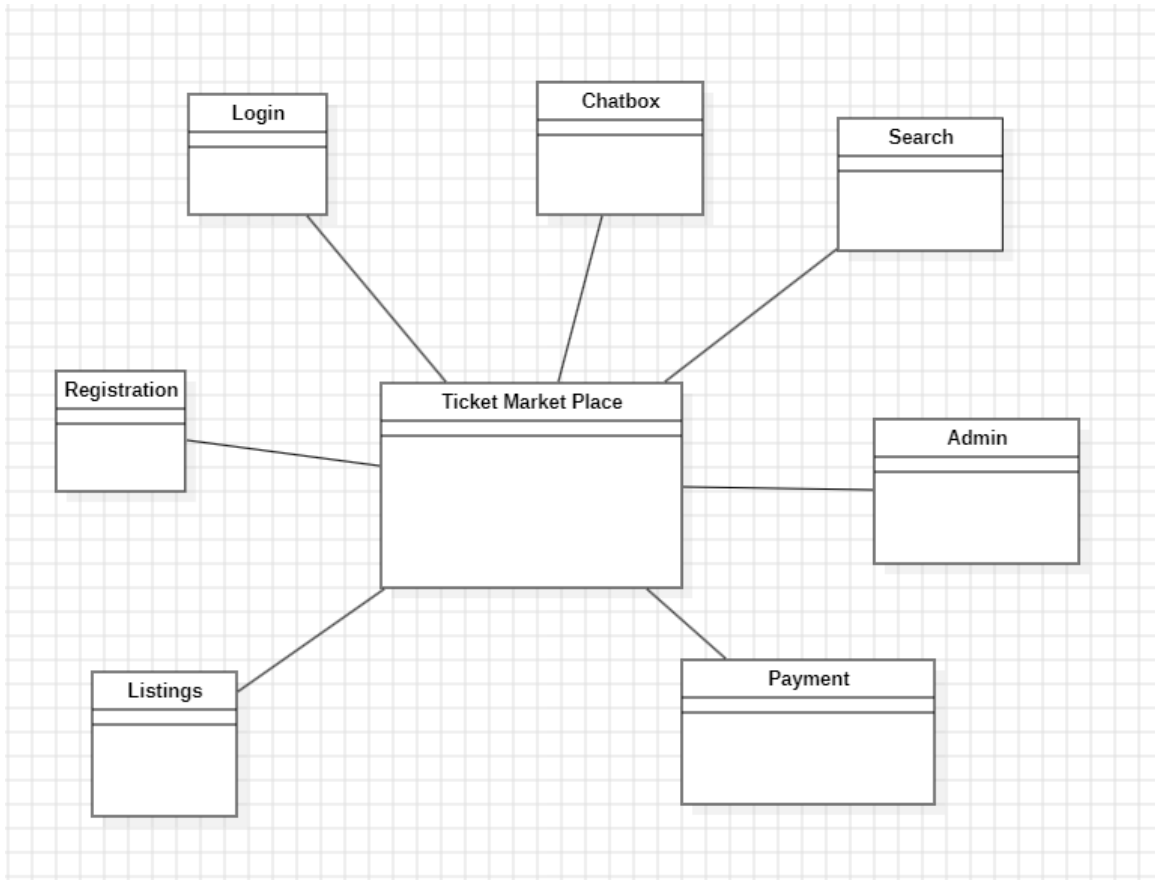
- Quantity
 - Ticket Price
 - Total Bill
 - Payment Method
 - Confirmation Message
 - Error Message
6. **Admin management:** The system must provide a management platform for administration that allows the admin to read reports, review, and delete fraud listings and scam users. This will include:
- Review user data
 - Review listing
 - Report list
 - Listing Removal
 - User Ban
7. **Chatbox:** The system provides a message platform that allows buyers and sellers message each other to reach a final agreement(optional):
- Sender ID
 - Receiver ID
 - Message Content

2.2 SYSTEM OBJECTIVES

1. The system will be android application for the mobile platform
2. The system will use email for account verification, notification, and transaction confirmation
3. The system will support in-app communication between buyers and sellers
4. The system will provide role-based access depending on the user role: buyer, seller or admin
5. The system will require an internet connection to enable real-time ticket browsing, listing transaction and messages.
6. The user interface will be intuitive, and designed to support navigation across major system features.

3. Project Context Diagram

The following is a context diagram displaying all high-level components of the system.



4. Systems Requirements

The following section breaks down each objective by describing the requirements needed in order to fully understand the functionality of the system. This focuses on the user functionality and does not include information regarding the hardware, database, test data, or android versions.

4.1 “LOGIN” REQUIREMENTS

Requirement Title:	Login
Sequence No:	1.1
Short description:	Must be able to correctly accept username and password logging in users
Description:	Take a Username or an email and Password from user, compare with the database of users and give the user access to their account.
Pre-Conditions:	User has registered
Post Conditions:	User is logged in has as access to their account
Other attributes:	Must be secure, a way to reset the password for the account High priority

4.2 “REGISTRATION” REQUIREMENTS

Requirement Title:	Registration
Sequence No:	1.2
Short description:	

	Must be able to take a new username, password, and email to add a new user to the database.
Description:	User inputs a new username, password, and email, then gets a registration confirmation email for security reasons, where they confirm their registration,
Pre-Conditions:	The new username and email have not been used before
Post Conditions:	User is logged in has as access to their account
Other attributes:	Must be secure High priority

4.3 "SEACH" REQUIREMENTS

Requirement Title:	Search
Sequence No:	1.3
Short description:	User types in keywords to bring up listings that match those keywords.
Description:	Takes a key word and has options for simple filters such as price, then creates a list of all listings in the database that match those keywords and filters.
Pre-Conditions:	The user has logged in.
Post Conditions:	User can now view the listings that they requested
Other attributes:	Efficient search algorithm. High priority

4.4 “CREATE LISTING” REQUIREMENTS

Requirement Title:	Create Listing
Sequence No:	1.4
Short description:	Takes user input to create a new listing, and adds that to the database.
Description:	Takes images, description, title, and price for the user, adds all of that to the database
Pre-Conditions:	User is logged in.
Post Conditions:	A new Listing has been created and can be accessed by other user.
Other attributes:	Must connect the Listing to the User High priority

4.5 “ADMIN MANAGEMENT” REQUIREMENTS

Requirement Title:	Admin management
Sequence No:	1.5
Short description:	

	Allows an “admin” user to access all of the listings and users on the database and make any changes necessary.
Description:	Take a specific password and username that’s designated as admin; this user then logs in and can delete, edit, and anything else necessary to all users and listings.
Pre-Conditions:	Users have registered as an admin.
Post Conditions:	Users have control over users and listings.
Other attributes:	Must be secure Medium priority

4.6 “CHECKOUT & CART” REQUIREMENTS

Requirement Title:	Checkout & Cart
Sequence No:	1.6
Short description:	Users are able to add listings to the cart and can go to a separate screen to checkout.
Description:	Users have a button on Listings to add to their cart, and then have to ability to click on a button to checkout, this brings to a page where they can review their cart, add remove items, and click checkout which brings them to the home page as we don’t have a payment feature.
Pre-Conditions:	User is logged in, there are other listings
Post Conditions:	User has checked out
Other attributes:	Medium priority

4.7 "CHATBOX" REQUIREMENTS

Requirement Title:	ChatBox
Sequence No:	1.7
Short description:	On a Listing, there is a way to access a screen where you can chat with the seller in order to bring the price down
Description:	Somewhere on the listing page, or a user page, be able to create a chatbox with the seller. There should also be a seperate page with all of the other chats that you can create.
Pre-Conditions:	User has registered, and logged in, as well as verified there account via email.
Post Conditions:	The seller and customer can negotiate the price.
Other attributes:	Lowest priority

5. Software Processes and Infrastructure

Development Process

The project follows Waterfall-like process organized into feature-based sprints. Each iteration includes requirement selection, design refinement, implementation, testing, and documentation updates. A functional build is produced at the end of every sprint.

Each iteration consists of:

Requirement Definition – Selected requirements are clarified and assigned acceptance criteria.

UML Design Update – Relevant UML artifacts (e.g., sequence, activity, and class diagrams) are revised to reflect the selected features.

Implementation – Feature modules are developed using modular design principles.

Testing – Unit and integration tests are executed to verify functionality.

Review & Integration – Code and documentation are reviewed prior to merge.

Planned iteration structure:

- Iteration 1: Authentication (Registration and Login)
- Iteration 2: Marketplace Listings and Search
- Iteration 3: Cart and Checkout
- Iteration 4: Payment Processing
- Iteration 5: Chat and Administrative Controls
- Iteration 6: System Integration and Final Validation

Testing Plan:

- **Unit Testing** – Verifies individual service logic (e.g., search filtering, order state transitions, payment validation).
- **Integration Testing** – Ensures proper interaction between components (e.g., checkout and payment workflow).
- **System Testing** – Validates complete application behavior against functional requirements.

Version Control Plan:

The project uses GitHub for configuration management.

Branching structure:

- **main** – Stable release branch
- **develop** – Integration branch
- **feature/<feature-name>** – Feature-specific branches

Defects are documented and resolved within the version control workflow before release.

Configuration and Traceability

Traceability is maintained by linking requirements, UML artifacts, implementation modules, and test cases.

When a requirement is modified:

1. Documentation is updated.
2. Relevant UML diagrams are revised.
3. Code is modified accordingly.
4. Associated tests are updated.

5.1 HARDWARE AND INFRASTRUCTURE

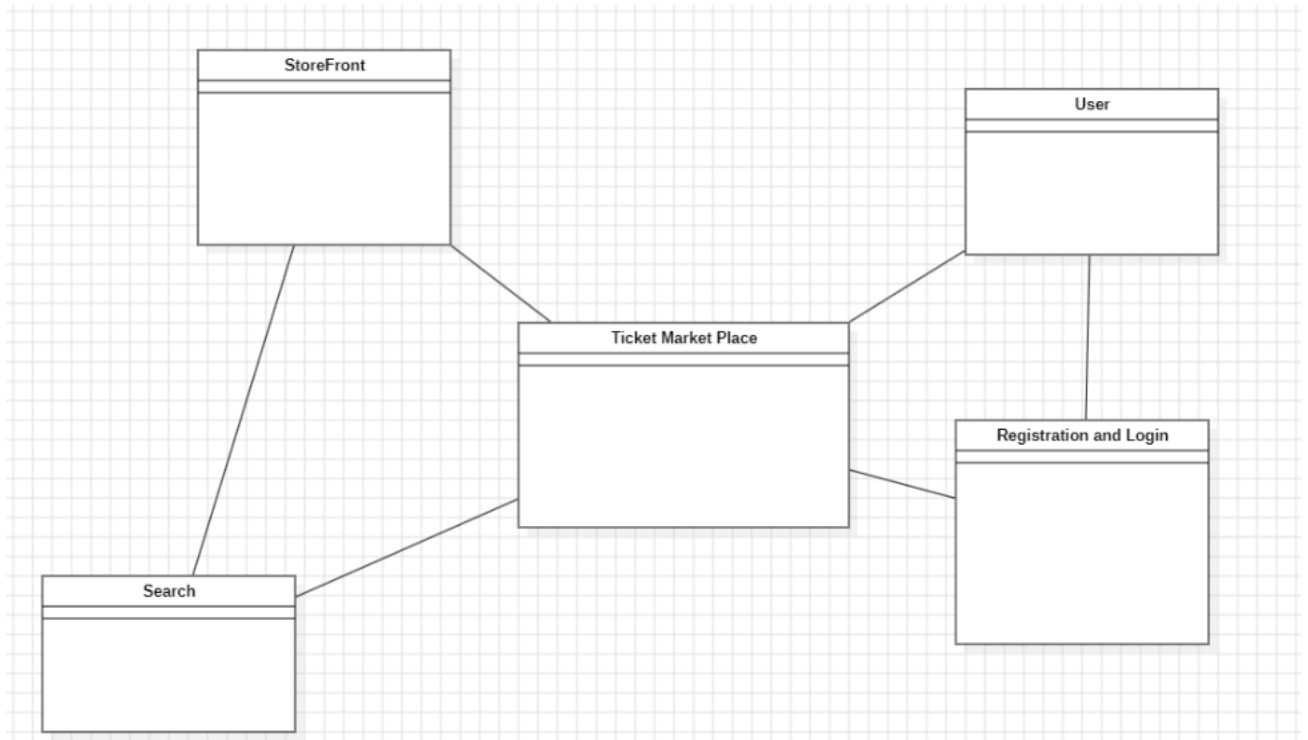
Android version:

The application targets **Android 10 (API Level 29) and above** and is designed to run on modern Android smartphones with internet connectivity.

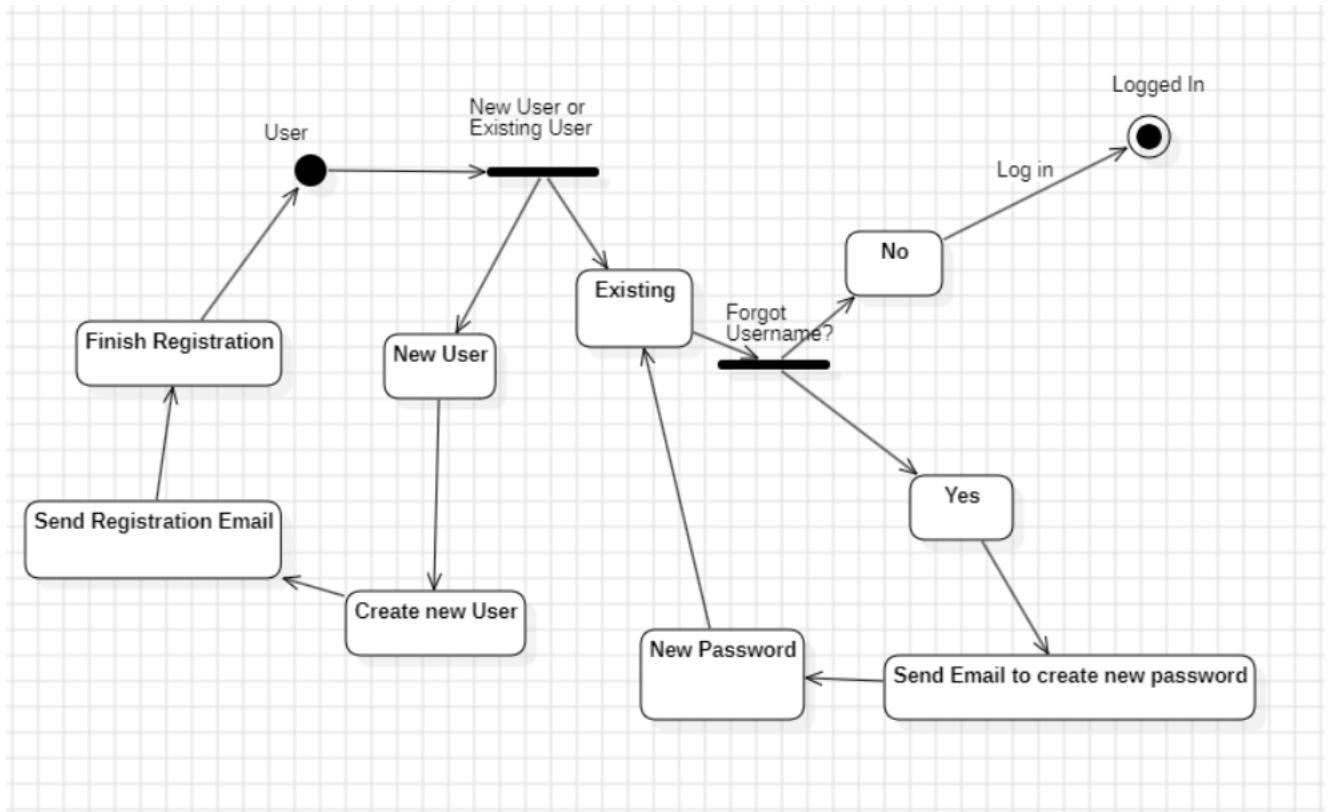
Android development environment:

The application is developed using **Android Studio** with the Android SDK and Gradle build system. Git is used for version control. Core marketplace features (search, checkout, and payment) are implemented within the application. Payment functionality uses a **mock payment gateway** for simulation. Data is stored using local persistent storage (e.g., SQLite).

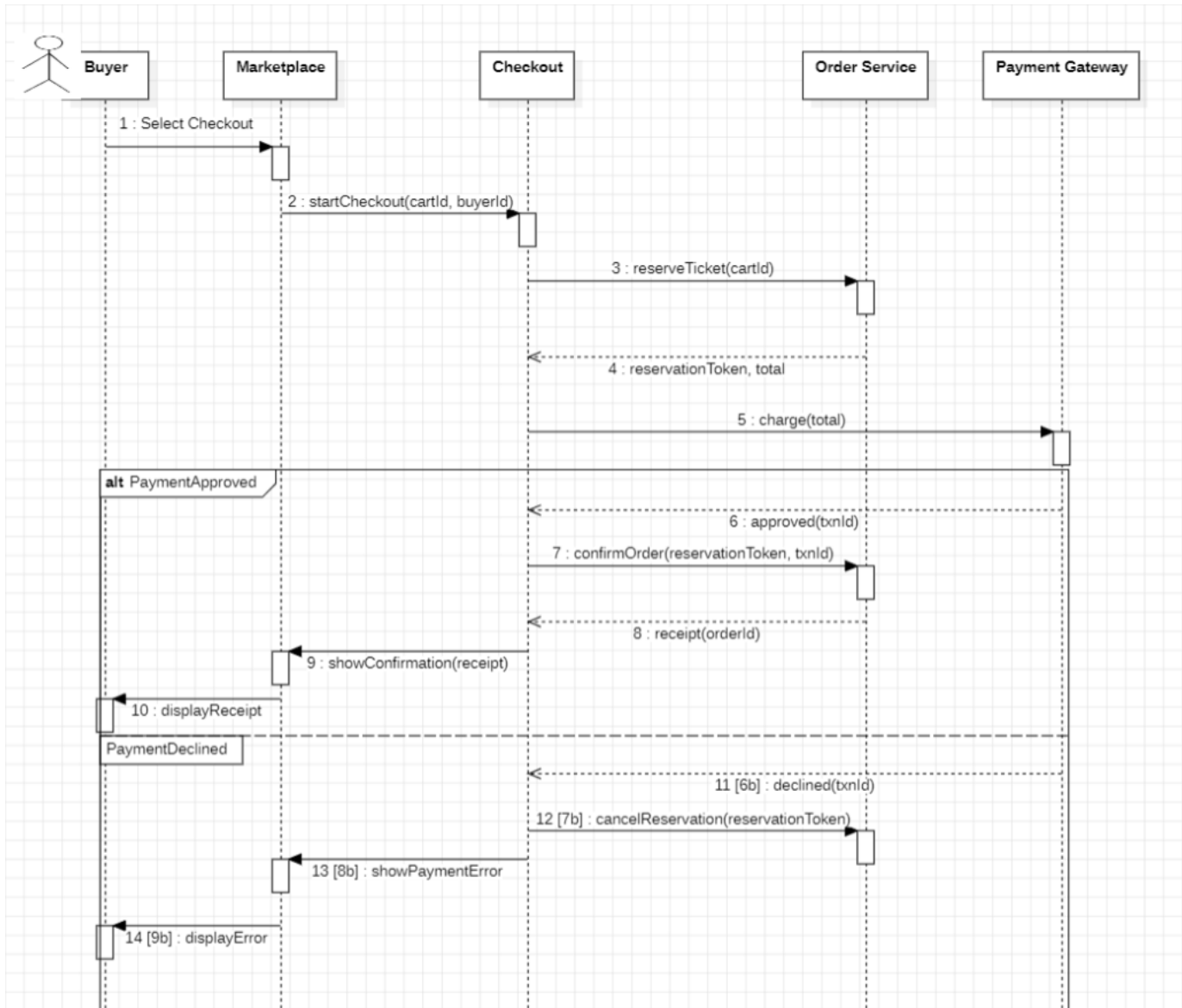
5.2 UML DIAGRAM: “SYSTEM”



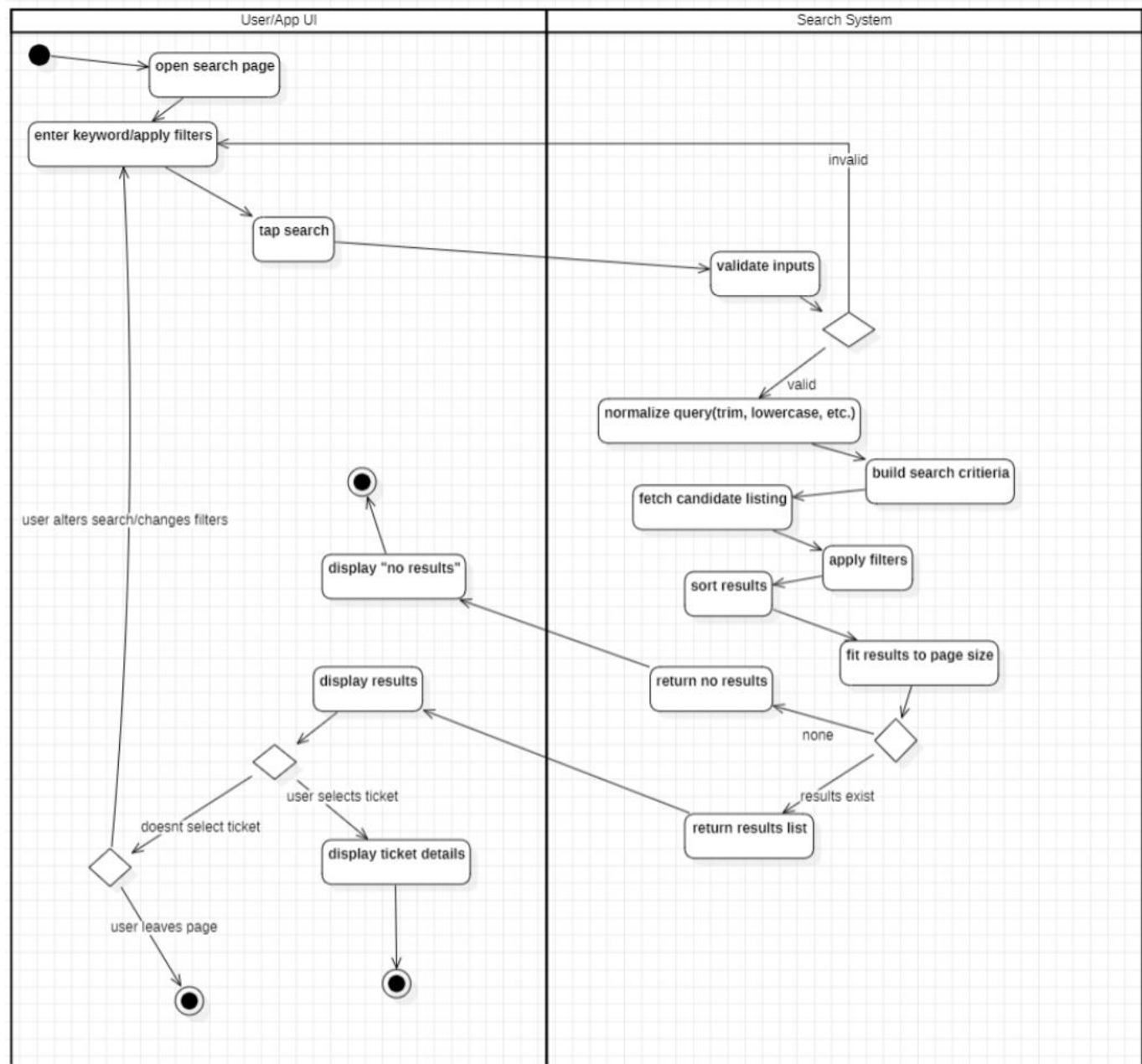
5.3 UML DIAGRAM: "REGISTRATION AND LOGIN"



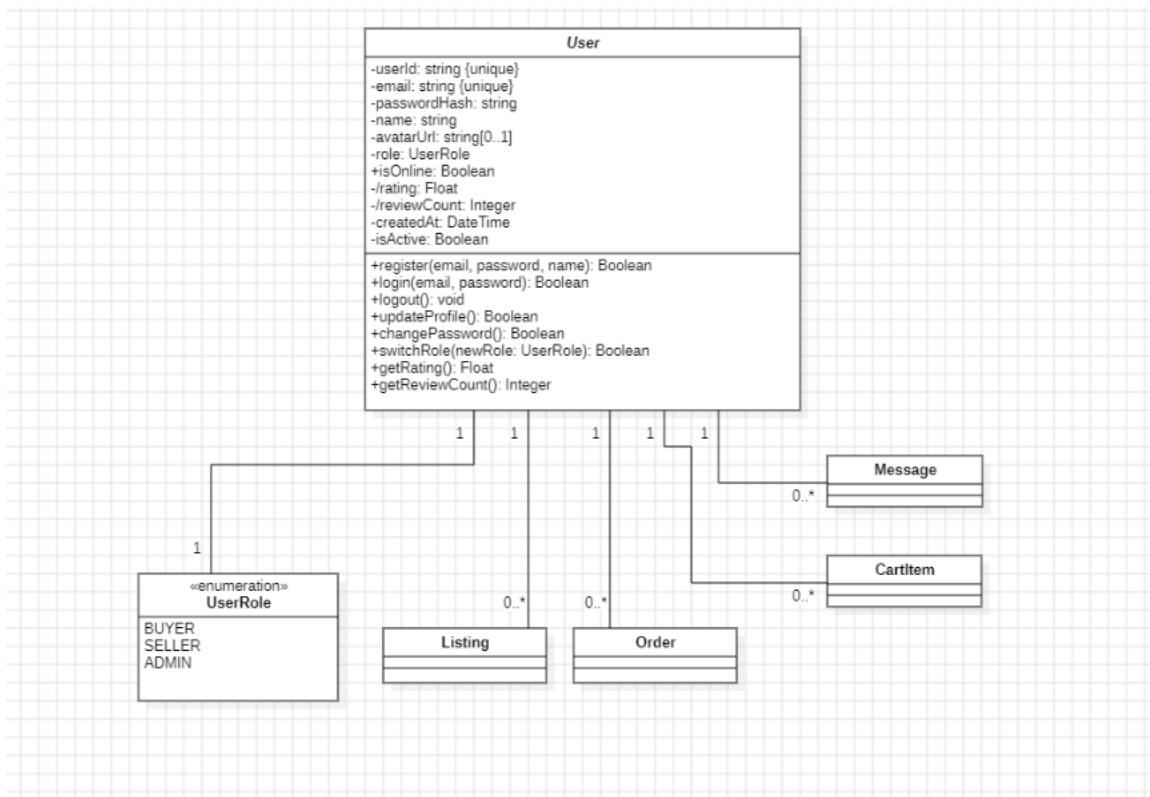
5.4 UML DIAGRAM: "PAYMENTS"



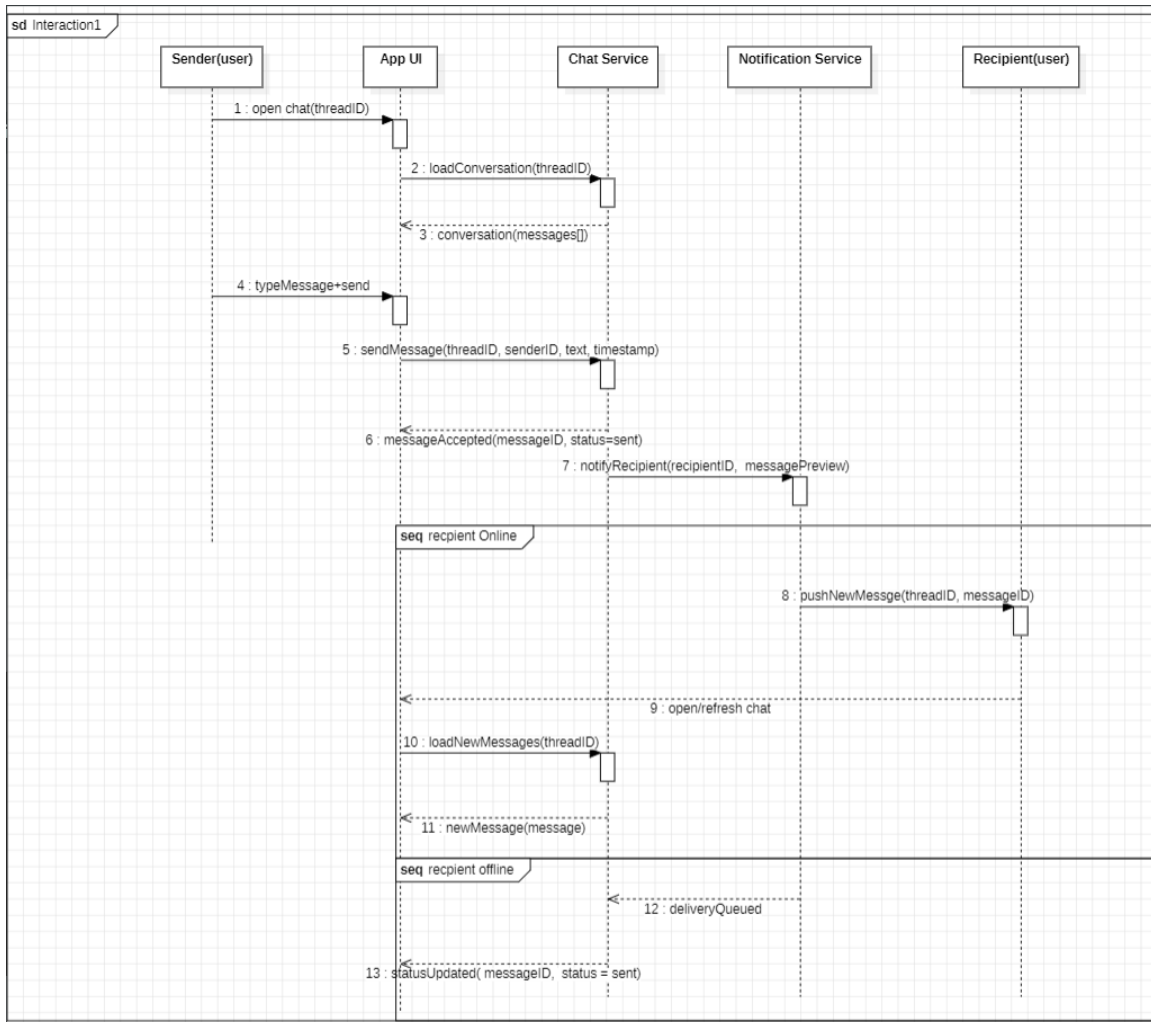
5.5 UML DIAGRAM: “SEARCH”



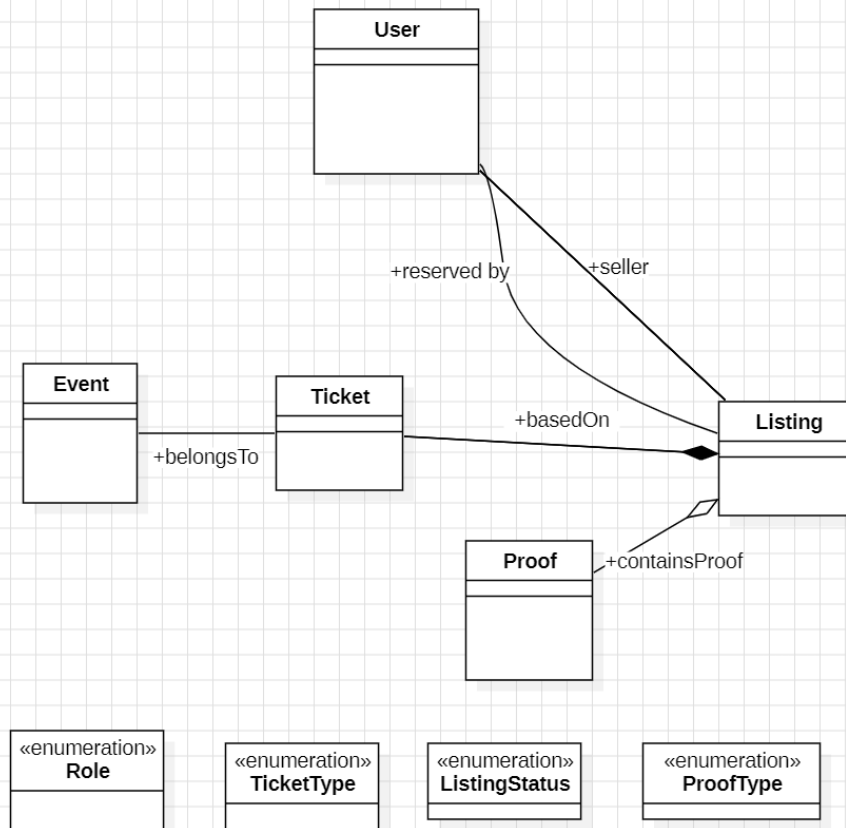
5.6 UML DIAGRAM: “USER”



UML Diagram: 5.7 “Chatbox”



UML Diagram: 5.8 “Listings”



6. Assumptions and Constraints

6.1 ASSUMPTIONS

1. **Internet Availability:** All users will have a stable internet connection when accessing the application, since real-time ticket browsing, messaging, and checkout require network connectivity.
2. **User-Provided Data Accuracy:** Users are assumed to provide accurate and truthful information during registration and ticket listing creation.
3. **Ticket Ownership Validity:** Sellers are assumed to legally own the tickets they upload and provide valid proof of ownership (physical image or digital confirmation).
4. **Administrative Oversight:** It is assumed that administrators will actively monitor user reports and review suspicious listings within a reasonable timeframe (e.g., within 24 hours).
5. **Email Accessibility:** Users will have access to a valid email account for verification, password recovery, and transaction confirmation.
6. **Platform Usage Scope:** The application will be used primarily for academic and prototype demonstration purposes, not for large-scale commercial deployment.

6.2 CONSTRAINTS

1. **Platform Constraint:** The system will be developed exclusively as an Android mobile application.
2. **Mock Payment Limitation:** The checkout process will simulate payment functionality. No real financial transactions or external payment gateway integration will be implemented.
3. **Time Constraint:** The project must be completed within the academic semester timeline.
4. **Resource Constraint:** Development will be performed by a small student team with limited development resources.
5. **Security Scope Limitation:** Advanced fraud detection mechanisms (e.g., AI-based verification or third-party ticket validation) will not be implemented.
6. **Scalability Limitation:** The system is designed for moderate usage and is not optimized for enterprise-level scalability.

6.3 OUT OF SCOPE MATERIAL

1. Integration with real-world ticket vendors or event organizers.
2. Real payment processing (credit card, PayPal, etc.).

3. Automatic ticket authenticity verification through barcode scanning.
4. Cross-platform development (iOS or web application).
5. AI-based dynamic pricing or fraud detection.
6. Legal dispute resolution mechanisms between buyers and sellers.

7. Delivery and Schedule

List all tasks/milestones from start of the project to the end with specific dates for both Anticipated Start & End Dates

Task/Milestone Description	Anticipated Start Date	Anticipated End Date	Status	Comments
Prepare Requirements and UML diagram	01/28/2026	02/13/2026	Complete	
<u>Milestone:</u> UML Diagram Delivery		02/13/2026	Completed	Feedback Received
SRA document (Includes project objectives, Requirements and UML diagrams)	02/18/2026	03/02/2026	Complete	
<u>Milestone:</u> SRA Document Delivery		03/02/2026	Completed	Deliverable will be seen in the project presentation
Database design	03/03/2026	03/15/2026	TBC	
Screen design	03/03/2026	03/20/2026	TBC	
Test case design	03/05/2026	03/31/2026	TBC	
External Documentation (i.e. User Manual)	03/10/2026	04/10/2026	TBC	
<u>Milestone:</u> Test Plan Delivery		04/01/2026	TBC	Deliverable will be seen in the project presentation
Project presentation	04/15/2026	04/24/2026	TBC	
<u>Final Milestone:</u> Project Delivery		04/24/2026	TBC	

8. Stakeholder Approval Form

Stakeholder Name	Stakeholder Role	Stakeholder Comments	Stakeholder Approval Signature and Date
Bhanu Jain	Developer Mgr		
Shaikh Islam	Project Assistance		
Keegan Boutwell	Developer		
Jake Milligan	Developer		
Phuc Nguyen	Developer		
Dogan Ogut	Developer		

Appendix:
